CSE 2500 - Introduction to Discrete Systems	Fall 2022
Assignment 3	Due date: Sep 29, 11:59pm, EST

For full credit, please adhere to the following:

- Unsupported answers receive no credit.
- All answers can be typed or handwritten, and should be readable.
- Submit the assignment in one file (.pdf, .doc, etc.) via HuskyCT.

Most of the following questions are in the textbook for the course, *Discrete Mathematics with Applications (5th Edition)*, by Susanna Epp.

1. Exercise Set 3.1
   (a) (5 points) Question 32 part (d)
   Let $\mathbf{R}$ be the domain of the predicate variable x. Which of the following are true and which are false? Give counter examples for the statements that are false
   (d) $x^2 > 4 \Leftrightarrow |x| > 2$ true
   (b) (5 points) Question 33 part (d)
   Let $\mathbf{R}$ be the domain of the predicate variables a, b, c, and d. Which of the following are true and which are false? Give counterexamples for the statements that are false.
   (d) $a < b$ and $c < d \Rightarrow ac < bd$ false if exactly one of b and d is negative but both of a and c are negative the product of a and c will be greater than the product of b and d
2. (10 points) Exercise Set 3.2, Question 19
   Write a negation for the following statement.

   $\forall n \in \mathbf{Z}$, if n is prime then n is odd or $n = 2$. there exists some n where n is not prime n is not odd and n is not 2
3. (10 points) Exercise Set 3.2, Question 29
   Write the converse, inverse, and contrapositive. Indicate as best as you can which among the statement, its converse, its inverse, and its contrapositive are true and which are false. Give a counterexample for each that is false. converse - if d = 3 then 6/d is an integer - true inverse - if 6/d is not an integer then d does not equal 3 - true contrapositive - if d does not equal 3 then 6/d is not an integer - false if d = 2 then 6/d is an integer

   $\forall$ integer d, if $6/d$ is an integer then $d = 3$.
4. (10 points) Exercise Set 3.2, Question 47
   The computer scientists Richard Conway and David Gries once wrote:

   The absence of error messages during translation of a computer program is only a necessary and not a sufficient condition for reasonable [program] correctness.

   Rewrite this statement without using the words necessary or sufficient. Is the converse of this statement true?

The absence of an error message is not enough information to guarantee a program works properly
converse = it is only necessary and not sufficient for program correctness the absence of an error message

(I do not understand how I am meant to say if this is true or not)

5. (10 points) Exercise Set 3.3, Question 12
Let $D = E = \{-2, -1, 0, 1, 2\}$. Write negations for each of the following statements and determine which is true, the given statement or its negation.

$\forall x$ in D, $\exists y$ in E such that $xy \geqslant y$ true x = 1 y = 2

there is some x in D, for all y xy < y
true when x is positive and y is less than 0
or when y is positive and x is 0

e <= 0 for all
sigma <= 0
such that x, if
a - sigma >= x
>= a +sigma
and x = a then
L - e >= f(x) >=
L+e

6. (10 points) Exercise Set 3.3, Question 43
The following is the definition for $\lim_{x \to a} f(x) = L$:
For all real numbers $\epsilon > 0$, there exists a real number $\sigma > 0$ such that for all real numbers x, if $a - \sigma < x < a + \sigma$ and $x \neq a$ then $L - \epsilon < f(x) < L + \epsilon$. Write what it means for $\lim_{x \to a} f(x) \neq L$. In other words, write the negation of the definition.

7. (10 points) Exercise Set 3.3, Question 56
Let $P(x)$ and $Q(x)$ be predicates and suppose D is the domain of x. For the statement forms in each pair, determine whether (a) they have the same truth value for every choice of $P(x)$, $Q(x)$, and D, or (b) there is a choice of $P(x)$, $Q(x)$, and D for which they have opposite truth values.

p(x) q(x)
1 1
1 0
0 1
0 0

a b
1 1
0 0
0 0
0 0

a b

$\exists x \in D, (P(x) \wedge Q(x))$, and $(\exists x \in D, P(x)) \wedge (\exists x \in D, Q(x))$

they have the same truth tables because you can simply factor the equations to make them the same
there is no point in which they have opposite truth tables

8. (10 points) Exercise Set 3.4, Question 18
Some of the arguments in 7-18 are valid by universal modus ponens or universal modus tollens; others are invalid and exhibit the converse or the inverse error. State which are valid and which are invalid. Justify your answers.

- If an infinite series converges, then its terms go to 0. P->Q
- The terms of the infinite series $\sum_{n=1}^{\infty} \frac{n}{n+1}$ do not go to 0. R
- $\therefore$ The infinite series $\sum_{n=1}^{\infty} \frac{n}{n+1}$ does not converge. W

isInvalid = (~p+q)(r)(~w)
isInvalid = ~p~wr+qr~w
isInvalid when
w=0
r=1
p = 0 or q = 1

9. (10 points) Exercise Set 3.4, Question 22
Indicate whether the argument below is valid or invalid. Support your answers by drawing Venn diagram(s).

go to the bottom of the page

1 • All discrete mathematics students can tell a valid argument from an invalid one.
2 • All thoughtful people can tell a valid argument from an invalid one.
3 • $\therefore$ All discrete mathematics students are thoughtful

10. (10 points) Answer the following questions:

(a) $P(x) = (x^3 == x)$ with domain Z^+. Is 1.5 in the domain of the predicate variable x? Is 3 in the domain of the predicate variable x?

1.5 and 3 are not in the predicate values for x in p(x)
1.5^3=3.375 != 1.5
3^3=27 != 3

b)
q(x) -> p(x)
this is true for all positive integers since the domain is positive integers the absolute value bars do not matter and p(x) is simply saying x^2=x^2 which is always true when x>=0 which is what q(x) says

(b) Let $P(x) = (|x| * |x| == x^2)$ and $Q(x) = x \geqslant 0$. Consider the following statement $P(x) \rightarrow Q(x)$. Is the converse of this statement true for $x \in Z^+$?

(c) Let $P(x) = (x + |x| == 2x)$ and $Q(x) = (x \neq 0)$. Let the domain of x be the set of all integer imaginary numbers i.e., $\{i, 2i, 3i, ...\}$. Is the statement $Q(x) \rightarrow P(x)$ valid?

(d) Are "pre-baked" arguments always correct? If not, give an example of Modus Tollens that is logically valid, but not a correct argument.

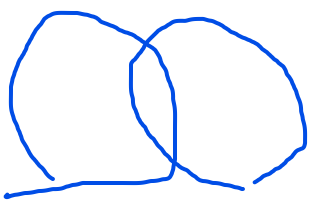
c)
the statement q(x)->p(x) is invalid
it is invalid because the set of imaginary numbers is close under addition but not closed under multiplication, so when evaluating p(x) it will not be true for all values of x when x is not equal to 0

d)
pre-baked arguments are always correct

9

1

2



not true the overlap is only a small portion of the statement